ASSIGNMENT-1

1.Dataset Description:

The California Housing Dataset contains information collected from California census districts. The dataset consists of 20,640 records and 8 input features.

Features Used

- MedInc – Median Income
- HouseAge – Median House Age
- AveRooms – Average Number of Rooms
- AveBedrms – Average Number of Bedrooms
- Population – Population of the Area
- AveOccup – Average Occupancy
- Latitude – Latitude Coordinate
- Longitude – Longitude Coordinate

Target Variable

- MedHouseVal – Median House Value

2. Exploratory Data Analysis (EDA)

EDA was performed to understand the structure and characteristics of the dataset.

Data Inspection

- The dataset contains 20,640 entries.
- All features are numerical.
- No missing values were present.

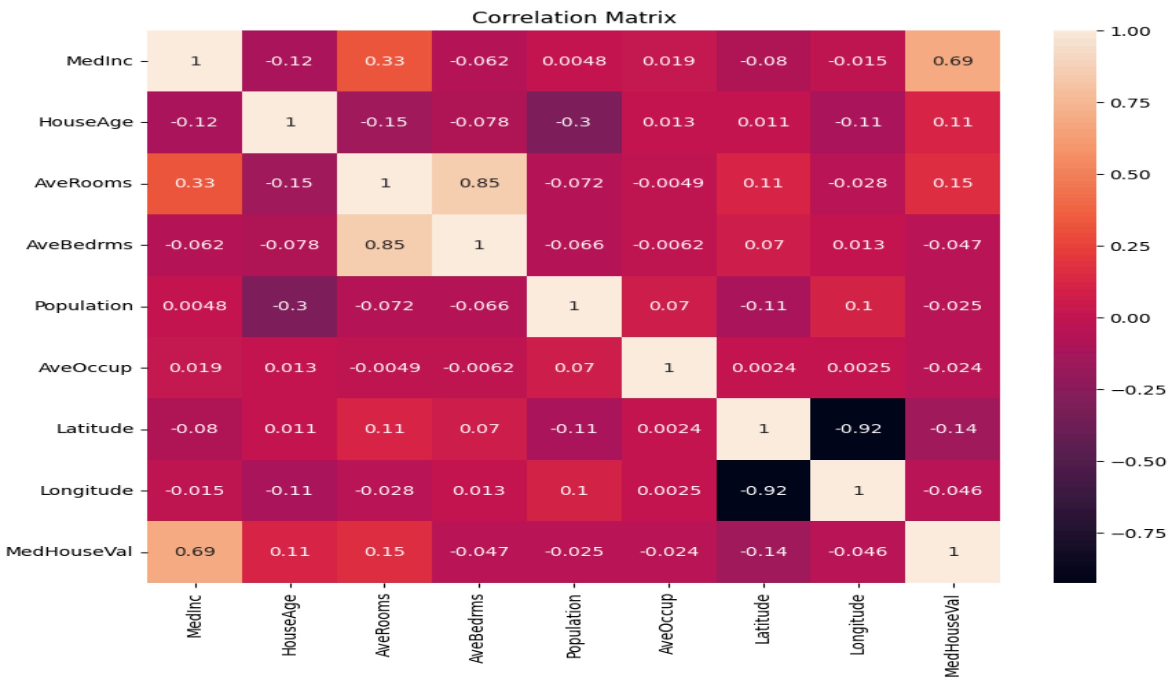
Correlation Analysis

A correlation heatmap was generated to analyze relationships between features and the target variable.

Correlation Heatmap Observations

- Median Income (**MedInc**) shows the strongest positive correlation with house prices (**0.69**), making it the most important predictor.
- Average Rooms (**0.15**) and House Age (**0.11**) have weak positive correlations with house prices.
- Latitude (**-0.14**) has a weak negative correlation with house prices.
- Population, Average Occupancy, and Average Bedrooms have very weak correlations, indicating limited individual influence on housing values.
- A strong positive correlation (**0.85**) exists between Average Rooms and Average Bedrooms, showing these features are closely related.
- Latitude and Longitude exhibit a strong negative correlation (**-0.92**), reflecting geographical dependence within the dataset.
- Overall, house prices are influenced by multiple factors, with **Median Income** having the **greatest impact.**

These observations indicate that income level is the most influential factor affecting housing prices in the dataset.



4. Model Development

The dataset was divided into training and testing subsets using an 80:20 split ratio.

Steps Performed

1. Load California Housing Dataset.
2. Perform exploratory data analysis.
3. Separate input features and target variable.
4. Split data into training and testing datasets.
5. Train a Linear Regression model.
6. Generate predictions on test data.
7. Evaluate model performance using MAE, RMSE, and R^2 score.

The Linear Regression algorithm was selected because it is simple, interpretable, and suitable for regression problems involving continuous outputs.

5. Results and Evaluation

The trained model was evaluated using standard regression metrics.

METRIC	VALUE
MAE	0.533
RMSE	0.746
R^2 Score	0.576

Interpretation

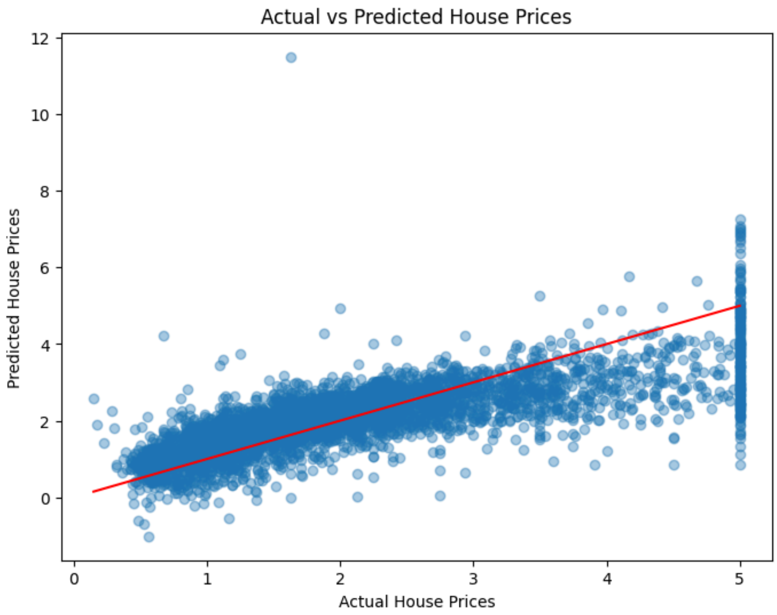
- The **MAE** value of **0.533** indicates that the average prediction error is relatively low.
- **RMSE** of **0.746** suggests that larger prediction errors are limited but still present.
- The **R^2 score** of **0.576** means that approximately 57.6% of the variation in house prices is explained by the model.

Overall, the model demonstrates moderate predictive performance and successfully captures the general trend in housing prices.

6. Visualization Results

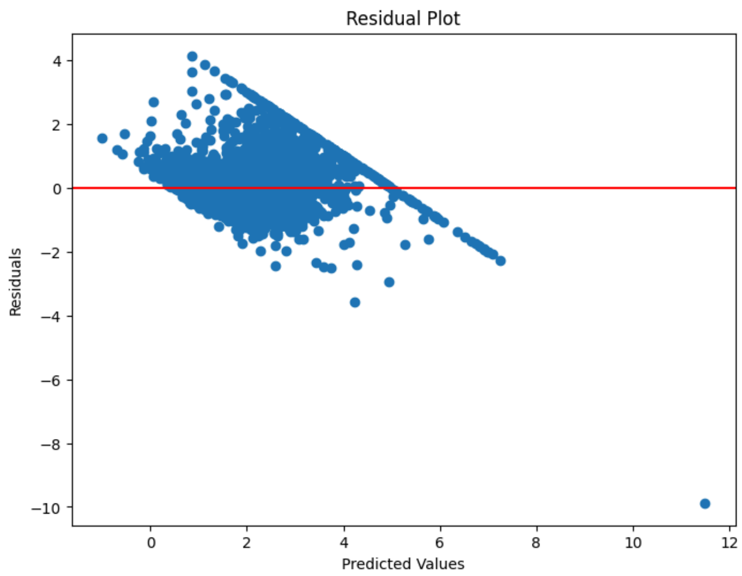
Actual vs Predicted Plot

The scatter plot comparing actual and predicted values shows a positive relationship between the two variables. Most data points are clustered around the ideal prediction line, indicating that the model captures the overall pattern of housing prices.



Residual Plot

The residual plot displays the difference between actual and predicted values. Most residuals are distributed around zero, indicating that the model predictions are reasonably unbiased. However, some patterns and outliers suggest that the relationship between features and house prices may not be perfectly linear.



7.Future Improvements

- Apply feature engineering and data preprocessing techniques such as outlier handling and feature scaling to improve model performance.
- Perform hyperparameter tuning and cross-validation to obtain more reliable and optimized results.
- Explore advanced regression models such as: Decision Tree Regression,Random Forest Regression,Gradient Boosting Regression,XGBoost Regression,Support Vector Regression (SVR),Polynomial Regression
- Incorporate additional real-world factors such as location amenities, transportation facilities, and economic indicators to improve prediction accuracy.
- Develop a simple web-based interface to enable real-time house price predictions for users.